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Modeling Self-Organization With Nonwellfounded Set Theory

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In this article, I use graphical tools from nonwellfounded set theory to model a variety of complex systems. I focus in particular on autocatakinetic systems. First there is a tutorial, explaining how to use the graphical tools of nonwellfounded set theory to model complex systems. I use these graphical tools to show that autocatakinetic systems are closed to efficient cause. Then I compare autocatakinetic systems with autopoietic systems. Finally, I use the graphical tools to examine autocatakinetic systems in light of Aristotle's 4 causes.

TUTORIAL

I begin by providing explicit instructions for using graphical tools from nonwellfounded (NWF) set theory. I preface these instructions with a brief explanation of the context in which Chemero and Turvey (2007a, 2007b, 2008; see also Chemero, 2008) introduced these tools to the study of complex systems.

NWF set theory (aka hyperset theory) was introduced as a tool for modeling phenomena that cannot be modeled with standard set theory (Aczel, 1988; Barwise & Etchemendy, 1987; Barwise & Moss, 1996). The main difference between standard and NWF set theory concerns the legality of circularity: in standard set theory, it is illegal for sets to contain themselves as members; in NWF set theory, it is legal for sets to contain themselves as members.

Following a suggestion by Kerckel (2003), Chemero and Turvey (2007a, 2008) used NWF set theory to model systems that are complex in the sense defined by Rosen (1991). Systems are complex in Rosen's (1991) sense when their models

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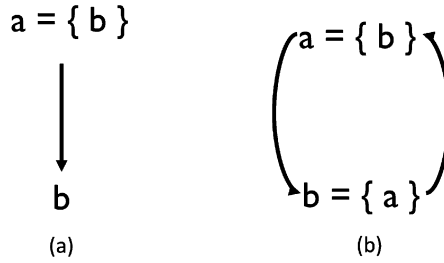


FIGURE 1 Nonwellfounded set graphs of (a) a predicative set and (b) an impredicative set.

contain *impredicativities* or circularities. Because NWF set theory is designed to deal with circularity, it is a useful tool for understanding systems that are complex in Rosen's (1991) sense. The main way that NWF set theory is useful for understanding complex systems is via its set graphing techniques. Consider a set $a = \{b\}$. To construct a graph of this set, draw an arrow from the set A to its sole member (Figure 1a). Now consider the sets $a = \{b\}$ and $b = \{a\}$. See Figure 1b. The graph of this set is a nonterminating loop. One way to understand the difference between standard and NWF set theory is in terms of graphs. Only graphs without terminating loops are allowed in standard set theory. Any graphable set is legal in NWF set theory.

Chemero and Turvey (2007a, 2007b, 2008) show that graphs of systems that are complex in Rosen's (1991) sense will have nonterminating loops. This implies two things. First, because graphs with nonterminating loops are illegal in standard set theory, only NWF set theory is sufficient to model systems that are complex in Rosen's (1991) sense. Second, one can use these graphs as complexity detectors. We also show that one can use these graphs to show that systems have another property important to Rosen (1991), "closure to efficient cause." A system is closed to efficient cause when all of the efficient causes of the system are also products of the system. In a graph of a system that is closed to efficient cause, all of the system's proper parts, excluding raw materials and waste products, will be involved in nonterminating loops. We see the graph of a system that is closed to efficient cause later.

With this as background, here are the steps to follow to build a graph of a system of interest.

Step 1

Think of the system of interest SI as a system of sets Σ . In general, there is no algorithm for generating the right Σ for SI . It will also typically be the case that there are many different Σ s that capture the causal structure of SI . In

earlier work, we have modeled *SI* (a) as functions, with mappings from inputs to outputs; (b) as collections of mappings from cause to effect; and (c) as collections of mappings from effects to causes. For the graphs presented here, I represent *SI* products as the set of entities that produce that set, that is, as $\text{Effect} = \{\text{Cause 1, Cause 2, } \dots \text{Cause N}\}$. For example, consider a very simple *SI* of (imaginary) autocatalytic reactions.



In this set of reactions, chemicals *R* and *S* are raw materials. If they are in steady supply, these reactions will form a self-sustaining network. We can represent this *SI* by equating each product with the set of reactants and catalysts that produce it. Thus the product *A* is $\{S, B\}$ and the product *B* is $\{R, A\}$.

Step 2

Graph the set. To do so, draw an arrow from each set to each of its members. Thus, one draws lines from $A = \{S, B\}$ to members *S* and *B* and from $B = \{R, A\}$ to members *R* and *A*. The result is Figure 2.

Step 3

Analyze the graph. If for any function *g*, chasing arrows from the function to its members, then to their members, and so on, leads back to *g*, the model Σ is impredicative and *SI* is *complex*. If every function of Σ is the output of another function, then *SI* is *closed to efficient cause*. Examining the graph of the autocatalytic set in Figure 2, it is clear that *SI* is both complex and closed to efficient cause. It is complex because there is at least one loop. It is closed to

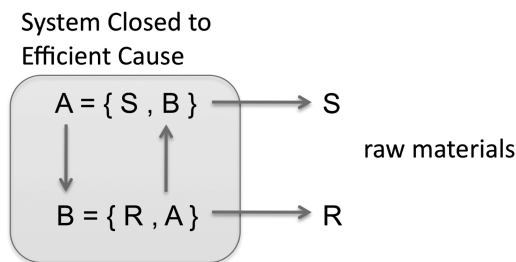


FIGURE 2 Nonwellfounded set graph of a simple autocatalytic system.

efficient cause because the parts of SI , A and B , are in a nonterminating loop; R and S are raw materials used by SI and so are outside the loop.

In the remainder of this article, I use the graphing tools from NWF set theory to examine autocatakinetic systems (Swenson, 1989; Swenson & Turvey, 1991), and to compare them to autopoietic systems (Maturana & Varela, 1980) and metabolism-repair systems (Rosen, 1991).

AUTOCATAKINETIC SYSTEMS, NONWELLFOUNDED SET THEORY, AND CLOSURE TO EFFICIENT CAUSE

It is important to note that Σ can be quite complicated without SI being complex in Rosen's sense. Figure 3 is a graph, a Σ , of such an SI , three cells of the cellular automata creature Bittorio (Varela, Thompson, & Rosch, 1991). This Σ graph is quite complicated but contains no nonterminating loops, so the depicted SI is not complex in Rosen's sense (see Chemero & Turvey, 2008, for discussion).

Some SI s that are complex in Rosen's sense are closed to efficient cause. As identified earlier, an SI is closed to efficient cause whenever all of its causal processes are the effects of other of its causal processes. These systems are self-organizing. That an SI is closed to efficient cause can also be demonstrated

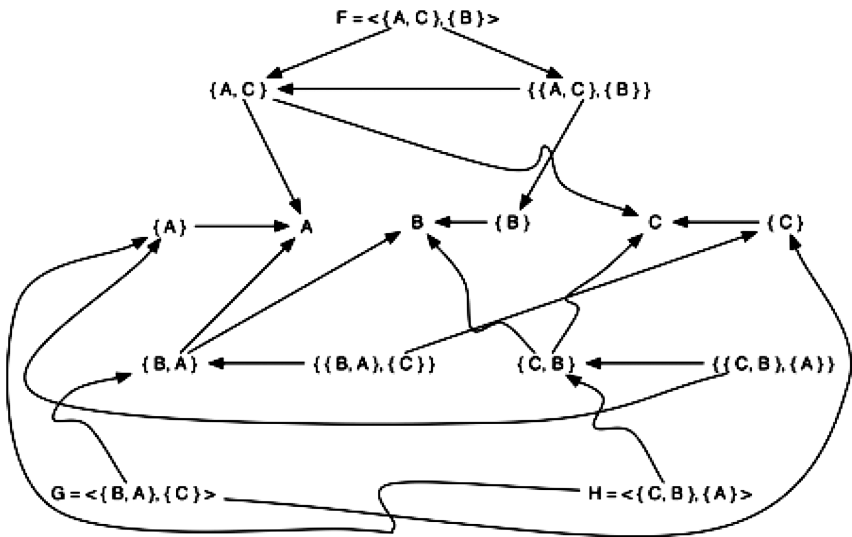


FIGURE 3 Nonwellfounded set graph of three cells of the cellular automata Bittorio, based on Varela, Thompson, and Rosch (1991).

graphically. Figure 4 is a graph of what Rosen (1991) calls a metabolism-repair *SI*. You can see in Figure 4 that each of the metabolism-repair system's functions (metabolism, behavior, and repair) is involved in a loop. The only terminal node in the graph is the raw materials the *SI* uses. Generally, we can say that an *SI* is closed to efficient cause whenever graphs of its Σ s are such that all of the system's causal processes are involved in nonterminating loops. In these graphs, the only terminal nodes are raw materials or waste products (i.e., things that are not properly speaking parts of the system). I proceed to show that autocatakinetic (ACK) systems have graphs in which all *SI* causes are involved in nonterminating loops and the only terminal nodes are raw materials and waste products.

A generalized ACK system (Swenson, 1989; Swenson & Turvey, 1991) is depicted in Figure 5. In this ACK system, the difference between an energy source E^I and an energy sink E^{II} generates a field potential with force F_1 and an ACK flow dE^I/dt . This ACK flow interacts with an internal energy source E^{III} to generate force with two results. First, there is a production of entropy, dS/dt . Second, an internal force F_2 is generated, and that internal force maintains the autocatakinetic flow dE^I/dt in an appropriate range. These two results of the interaction of the ACK flow with the internal potential require comment. The first result, the production of entropy, is in some sense the purpose of the ACK system according to the Law of Maximum Entropy Production

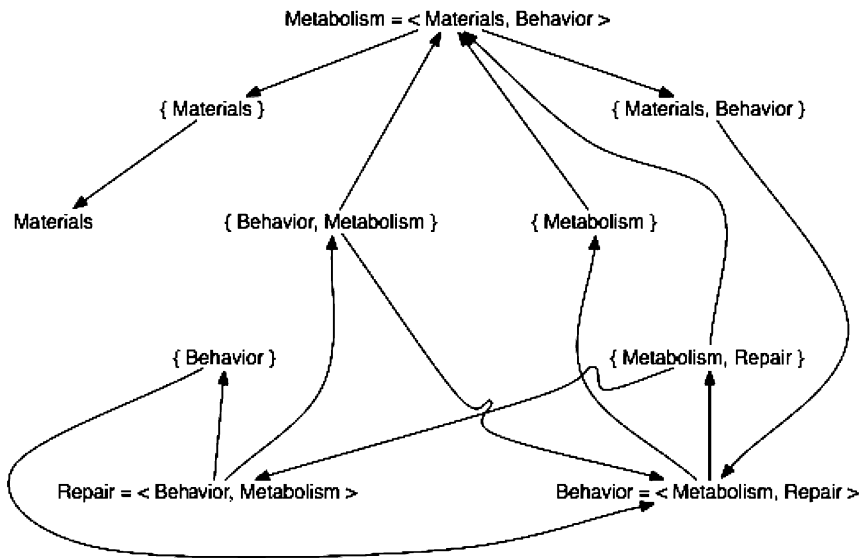


FIGURE 4 Nonwellfounded set graph of a metabolism-repair system, based on Rosen (1991).

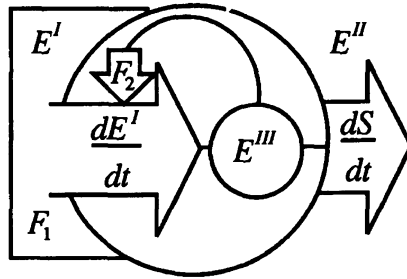


FIGURE 5 An autocatalytic system from Swenson and Turvey (1991).

(LMEP; Swenson, 1989; Swenson & Turvey, 1991). Second, the relationships among the internal potential E''' , the internal force F_2 , and the ACK flow dE'/dt are self-sustaining: the flow to the potential generates the force that maintains the flow to the potential.

Figure 6 is a graph of the generalized ACK system. To produce the graph, each entity generated by the system is depicted as the set of those entities that are involved in the system. This convention was also used in earlier applications of NWF set theory to understand complex systems, for example, with systems of autocatalytic chemical reactions in which each system product is taken to be the set of reactants involved in its production (Chemero, 2008; Chemero & Turvey, 2007a). Looking at the graph, one can see that the system is complex in Rosen's (1991) sense. There are loops (a) between E''' and F_2 and (b) connecting

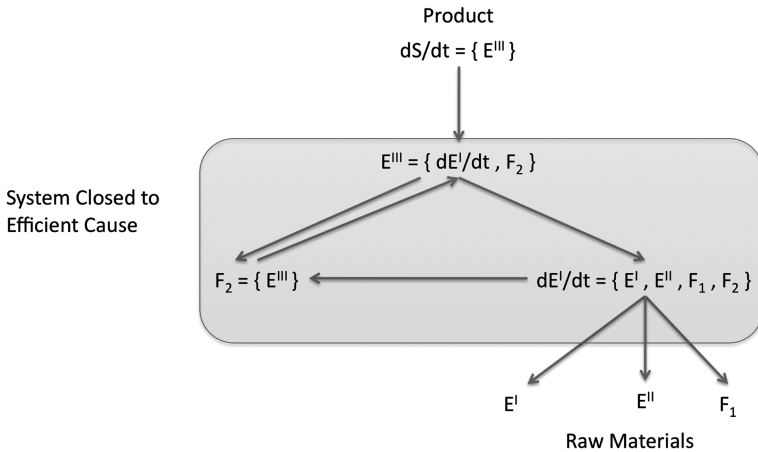


FIGURE 6 Nonwellfounded set graph of an autocatalytic system.

dE^I/dt , F_2 , and E^{III} . One can also see that the system is closed to efficient cause. The ACK flow dE^I/dt , the internal potential E^{III} , and the internal force F_2 form a closed network of causes, each of which is involved in nonterminating loops that include the others. The only terminal nodes of the graph are for system raw materials (E^I , E^{II} , and the force generated by their difference F_1). Moreover, the only thing produced by the system that is not part of the system itself is dS/dt , which is in perfect agreement with LMEP, according to which ACK systems exist because, for the given components and potentials, they maximize the rate of entropy production.

We can now see what ACK systems look like when depicted with the graphing tools from NWF set theory. By depicting other theoretical models of self-organization using the same tools and conventions, it is possible to compare them with Swenson's (1989) generalized ACK system. In particular, it is possible to see the ways in which ACK systems differ from autopoietic systems (Maturana & Varela, 1980) and metabolism-repair systems (Rosen, 1991). We can also use Figure 6 as an ACK detector. To do so, graph some candidate system, using the convention that a product is the set of its causes (product = {Cause 1, Cause 2, ... Cause N}). Then ask two questions: Does the graph contain a nonterminating loop that includes the maintenance of an environmental energy gradient? Does the graph have a term for entropy that has arrows coming from it but none going to it? If the answer to both questions is yes, the system in question is ACK.

AUTOCATAKINESIS AND AUTOPOIESIS

Earlier I used the graphical tools from nonwellfounded (NWF) set theory to show that autocatakinetic (ACK) systems are closed to efficient cause. In an earlier paper, Turvey and I (Chemero & Turvey, 2008) used the same tools to show that autopoietic systems are also closed to efficient cause. This is one way in which ACK and autopoiesis are similar. There are also important differences between these two types of system. Swenson (1992) indicted autopoietic theory on many counts (scientific, conceptual, epistemological) and argued that ACK is an appropriate foundation for ecological physics. Here I set most of Swenson's criticisms aside and focus on what Maturana and Varela (1980) called the most basic autopoietic entity: the cell. I use the graphical tools of NWF set theory to directly compare the general ACK system with the cell as understood in autopoietic theory.

Look again at Figure 6, a graphical depiction of that system using the graphical tools from NWF set theory. Because the system internal energy E^{III} , the system internal force F_2 , and the autocatakinetic flow dE_I/dt are in a nonterminating loop in Figure 6, we can conclude that these three elements form a system that is closed to efficient cause.

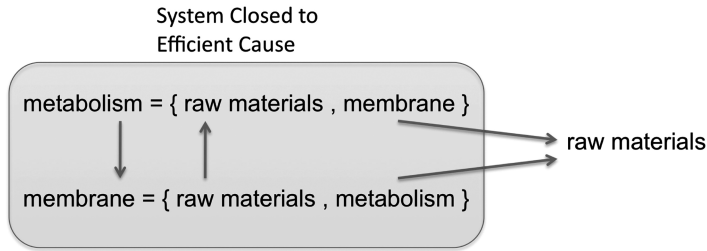


FIGURE 7 Nonwellfounded set graph of a cell, as understood in autopoietic theory.

Figure 7 is a graph of the cell as described in autopoietic theory. The autopoietic cell has two relevant components. First, there is its metabolism, which uses raw materials to maintain the other main component of the cell, the membrane. The cell membrane's main activity is to use environmental raw materials to bound the metabolism, making itself possible. Maturana and Varela (1980) say that the cell is an operationally closed system—the focus of the activity of each component is the other components of the system. The graph of the cell in Figure 7 uses the same conventions as the graph of the ACK set in Figure 6: each component of the system is depicted as the set of its causes. Notice that the metabolism and the cell membrane are in a nonterminating loop, so the cell is closed to efficient cause. In this way, the autopoietic cell and the generalized ACK set are similar. The remainder of this section focuses on the ways that they differ. In describing these differences, I refer to Figure 7 as a depiction of an autopoietic system rather than as a depiction of a cell. I do so because whether the autopoietic cell is an accurate description of actual cells is a matter of dispute: Swenson (1992) would claim that a cell is an ACK system, not an autopoietic one.

The first difference to be noticed is that the graph of the ACK system includes a product, that is, something that is generated by the system but is not part of the system; the autopoietic system does not include such a product. The product of the ACK system is increased entropy. This makes sense in the light of LMEP, according to which ACK systems exist because, for the given components and potentials, they maximize the rate of entropy production. That the autopoietic system has no depicted products might seem surprising in light of the claim by proponents of autopoiesis that autopoietic systems enact or “bring forth” worlds. The graph, however, gets things right. The graph depicts the system's causes, and the sense in which autopoietic systems bring forth worlds is not causal. It is hard to say precisely what bringing forth worlds amounts to, and philosophical terms of art like *constitute* are often used. Although this leads to the ultimately correct feeling that autopoiesis is a troublingly idealistic theory, proponents of

autopoiesis do not claim that the autopoietic systems are entirely walled off from the environment. For one thing, proponents of autopoiesis claim that autopoietic systems are “structurally coupled” to the environment. For another, Maturana and Varela (1980) would never deny that autopoietic systems cause changes to the environment; they just think that those changes are by-products and are not important for describing the nature of the system. In contrast, though, producing entropy in the environment (at the fastest possible rate given the conditions) is the function of ACK.

The second difference in the diagrams amplifies this sense that autopoietic systems are substantially more separate from the surrounding world than ACK systems are. Note that in Figure 7, the system that is closed to efficient cause (inside the rectangle) is composed of proper parts of the cell: the metabolism and the cell membrane. In contrast, the system that is closed to efficient cause in Figure 6 is the system internal energy E^{III} , the system internal force F_2 , and the ACK flow dE^I/dt . The ACK flow is the difference between the environmental energy source E^I and the environmental energy sink E^{II} , something that is not, intuitively speaking, contained within the system. In the ACK system, the environment is not merely the necessary backdrop but is part of the self-organizing system. ACK systems include their environment.

The graphs reinforce Swenson’s (1992) critique of autopoietic systems as inappropriate for ecological physics. Autopoietic systems interact with the physical environment but only as a source of raw materials. The world they experience is something separate and one that they “bring forth.” ACK systems are inseparable from the environment they act in, on, and for. This makes ACK, but not autopoietic, theory an appropriate foundation for ecological physics and physical intelligence. It would be easy to use the graphical tools of NWF set theory to show that Rosen’s (1991) metabolism-repair systems are more like autopoietic systems than like ACK systems in the respects described here. That is, metabolism-repair systems are also troublingly separate from their environments and are, therefore, also inappropriate for ecological physics and physical intelligence. (Remember: Rosen [1991] routinely makes idealist claims concerning the impossibility of access to the world beyond our models of it.) To do so, one should construct a graph of metabolism-repair systems that uses the same conventions as those used here. Then one could compare the metabolism-repair graph directly with those in this article. I leave this as an exercise for the reader.

AUTOCATAKINESIS AND ARISTOTLE'S FOUR CAUSES

We have seen that ACK, autopoietic, and metabolism-repair systems are closed to efficient cause. In this section, I focus specifically on ACK systems, examining

them in terms of Aristotle's three other causes. To begin this, it helps to briefly outline Aristotle's four causes. The first thing to note is that Aristotle was actually not talking about causes in the way that modern scientists do; instead, he was talking about explanations. So Aristotle's four causes are actually four ways of explaining some phenomenon. Imagine the following phenomenon: (a) some process makes (b) some material take (c) some form for (d) some purpose. (a)–(d) in the previous sentence refer, in order, to the efficient, material, formal, and final cause of this event. They are explanations of different aspects of the phenomenon. We can use graphical tools from category theory and NWF set theory to comment on each of these types of cause as they operate in ACK systems.

To begin, it helps to characterize the efficient causes in Swenson's generalized ACK system (Swenson, 1989; Swenson & Turvey, 1991) as a set of functions or mappings from inputs to outputs. Note that this is a more complex rendering of ACK systems than was used earlier and mirrors the way both Rosen (1991, 2000) and Chemero and Turvey (2007a, 2007b, 2008) depicted metabolism-repair systems. The basic idea is that a function is represented as an ordered pair with the first member being the set of inputs and the second member being the set of outputs. Thus,

$$\text{function} = \langle \text{inputs, outputs} \rangle .$$

Chemero and Turvey (2007a, 2007b, 2008) followed Rosen (1991) in understanding this as follows:

$$\text{efficient cause} = \langle \text{material cause, effect} \rangle .$$

The generalized ACK system, as described earlier, consists of three components densely interacting to maintain the environmental conditions appropriate for their continued existence and to generate entropy. To reiterate, this set of interactions or relations is captured in the following collection of functions:

$$E^{III} = \langle F_2, dE^I/dt \rangle$$

$$F_2 = \langle dE^I/dt, E^{III} \rangle$$

$$dE^I/dt = \langle E^{III}, F_2 \rangle, \langle F_1, dS/dt \rangle$$

These functions are depicted using diagrammatic tools from category theory in Figure 8. Figure 9 is a depiction of a metabolism-repair system using the same tools. In both figures, a solid arrow connects each function to its input and a dashed arrow connects the input to the output. Alternatively, one could say that each efficient cause is connected to a material cause with a solid arrow,

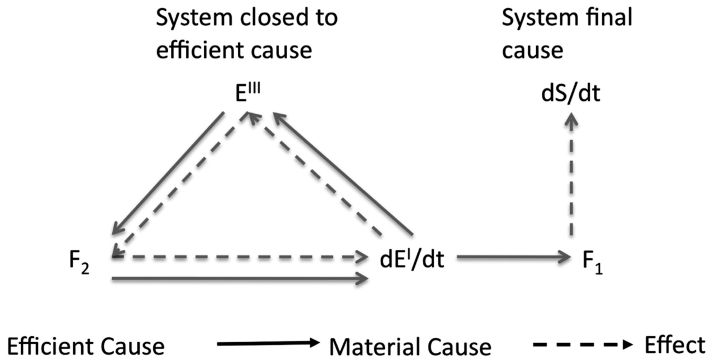


FIGURE 8 An autocatalytic system, depicted using conventions from category theory.

and a dashed arrow connects material cause to effect. Comparing Figures 8 and 9 is instructive. First, recall from the earlier discussion that metabolism-repair systems and the dE^I/dt , E^{III} , F_2 portion of the ACK system are closed to efficient cause. This is evident in Figures 8 and 9: each efficient cause (at the head of a solid arrow) is also an effect of the system (at the tail of a dashed arrow). But notice a key difference. In the ACK system, but not the metabolism-repair system, each component is efficient cause, material cause and effect of the other components in the triad dE^I/dt , E^{III} , F_2 . I call this feature the *ACK triangle*. Moreover, it is possible to see the final cause of the ACK system in Figure 7. The purpose or final cause of ACK systems is the creation of entropy. We can see in Figure 7 that the maintenance of the ACK flow dE^I/dt by the other components of the system allows the ACK system to use the environmental energy gradient to produce entropy. This entropy is a by-product of the activity of the system: in the course of self-maintenance, the ACK system generates entropy.

The same features of the ACK system are visible using the graphical tools of NWF set theory. We can depict the ACK system, understood as the set of

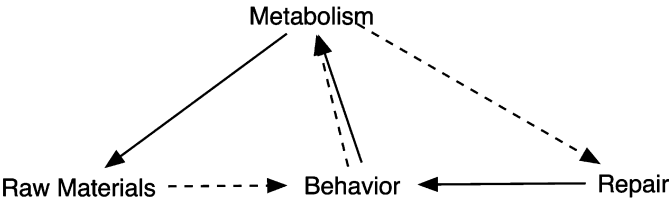


FIGURE 9 A metabolism-repair system, depicted using conventions from category theory.

functions described previously, if we use a common convention for rendering ordered pairs as sets. In an ordered pair, unlike in a set, the order of the members is significant. That is, $\langle a, b \rangle$ is different from $\langle b, a \rangle$, but $\{a, b\}$ and $\{b, a\}$ are the same. The convention in set theory is to depict ordered pairs as Kuratowski pairs, sets that include the first member of the pair as a singleton set and a set that includes both members of the ordered pair. Thus, the Kuratowski pair for $\langle a, b \rangle$ is $\{\{a\}, \{a, b\}\}$. Figure 10 is the NWF set theory graph of the functions of a generalized ACK system. As with Figure 7, the system divides naturally into two connected subsystems: the ACK triangle (which looks less triangular here) and the entropy generator. Notice in particular that the lower part of the graph, which contains the entropy generating components, has arrows going to it from other parts of the system but not leaving it. Entropy generation is the final cause of the system, and the entropy generated is not fed back into the system.

So far, I have said nothing about formal cause. It is tempting to take the ACK triangle as indicating something about formal cause in ACK. Each of the components of the triangle has its form, is the sort of thing that it is, in virtue of its efficiently causing, being material cause of, and being caused by the

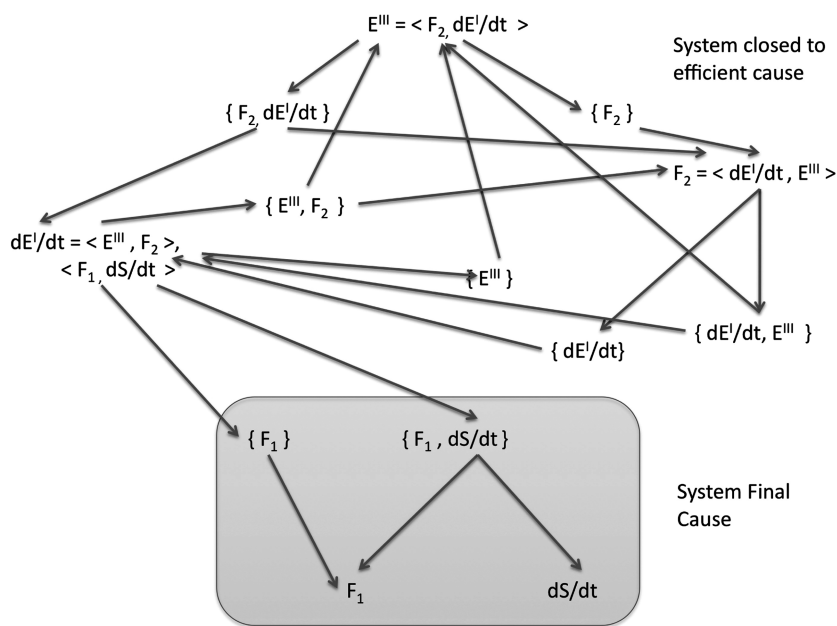


FIGURE 10 Nonwellfounded set graph of an autocatakinetic system, using Kuratowski pairs to depict functions.

other components of the triangle. Then the fact that the components have their particular form can be used to explain their behavior. As Kuhn (1977) points out, in one way or another, formal causation has been the dominant style of causal explanation in physics ever since Newton. For example, to explain the current position of Mars, one averts to the kind of thing Mars is: a body in a two-body Newtonian system with particular boundary conditions. More generally, explanation of the behavior of some entity in physics is given in terms of its form. Similarly, we might use the fact that entities play particular roles in an ACK system to explain their behavior. For example, consider Carriveau's (2006) discussion of hydraulic vortices as ACK systems. In his analysis, he identifies elements of his experimental setup as components in an ACK system as seen in Figure 11. In analogy with the way the behavior of Mars is explained by reference to its form (as a body in a two-body system), we might use the identification of the flow into the intake as dE^I/dt , the force due to gravity as E^{III} , and the stretching of the vortex as F_2 to explain the way the flow, the force, and the stretching interact over time. Or we might use the identification of these elements as components in an ACK system to explain the way in which entropy is produced by the system. The explanatory leverage provided by this use of formal cause to explain the behavior of systems that are identified as ACK is still to be determined.

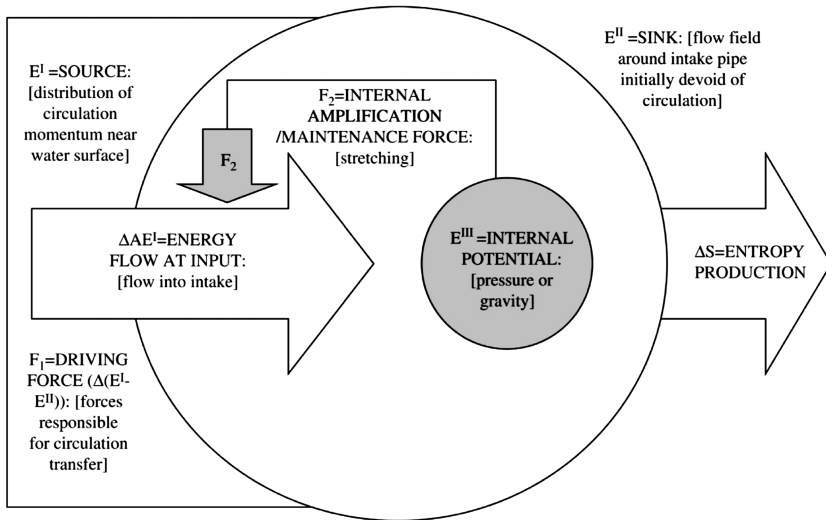


FIGURE 11 Hydraulic vortices as autokinetic systems, based on Carriveau (2006). Figure by Michael Turvey.

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